Criteria for Evaluating AGI Claims:
A Triptych of Logical, Operational, and Phenomenological Tests

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Abstract

Recent progress in artificial intelligence has led to increasingly strong claims about imminent artificial general intelligence (AGI)[1]. At the same time, empirical and theoretical work on scaling laws, out-of-distribution generalization, epistemic uncertainty, embodied intelligence, and machine consciousness has exposed important limitations in current computational approaches. Building on this literature, we propose a set of *architecture-independent criteria for evaluating AGI claims*, applicable to any system that purports to exhibit general intelligence or consciousness.

First, we formulate a *logical–computational consistency criterion*: scaling-hypothesis assumptions must be compatible with known constraints from neural scaling laws and compute trends [1–9]. We show that common informal arguments for "AGI via scaling alone" implicitly require implausible asymptotic behavior in energy, test-time compute, and sample efficiency, given information-theoretic limits [10] and recent analyses of emergent abilities [11–14].

Second, we introduce *operational engineering criteria* that any system plausibly claiming general intelligence should satisfy: (1) generating its own priors from embodied interaction [15–20], (2) tracking epistemic uncertainty rather than only point estimates [21–25], (3) robust out-of-distribution generalization [26–29], (4) intrinsic motivation and active inquiry [30–34], and (5) grounding in an updatable sensorimotor loop [35–38]. We argue that currently deployed systems do not yet robustly satisfy these criteria, and that proposed fixes (e.g., tool use, light embodiment, agentic scaffolding) only partially address them.

Third, we propose *phenomenological and moral-patienthood criteria* for AGI claims that invoke consciousness or sentience. We operationalize these via a mapping between classical Abhidharma mental factors (cetasikas) and contemporary indicators of machine consciousness [39–45], as well as Buddhist-inspired analyses of mind and care in AI [46–50]. This mapping (Figure 3) distinguishes perceptual–cognitive capacities that some current systems approximate from hedonic tone, vitality, and moral emotions, for which no non-simulated implementation is yet evidenced.

We illustrate the rubric using representative AGI-era claims and analyses from the early 2020s [51–58][2]. While the figures illustrate the present capabilities of contemporary machine-learning models relative to these criteria, the criteria themselves are architecture-independent and intended as falsifiable constraints on any future AGI claim.

We invite both proponents and skeptics of AGI to treat these criteria as falsifiable constraints on rhetoric, experiment design, and system characterization, rather than as a priori arguments for or against the possibility of machine general intelligence.

[1] At frontier scales, a single training cycle can consume tens of gigawatt-hours of electricity—the daily output of a small city—while 800 V accelerator racks demand cooling capacity better measured in fire-hydrants-per-rack than in laboratory units. These physical costs are not cosmetic; they mark the point where optimistic scaling curves collide with thermodynamic and engineering reality, effectively burning civic-scale infrastructure to chase marginal log-likelihood gains.

[2] The early-2020s cycle provides a convenient reference case: a period in which AGI rhetoric often outpaced evidentiary support, with claims grounded more in scaling optimism than in demonstrated advances in reasoning, grounding, or diachronic continuity. We treat this era only as an illustrative case study; the criteria themselves are intended to outlast any particular technological cycle.

Preface

This paper examines a tension at the center of contemporary AI discourse: the contrast between linguistic fluency and actual intelligence.[3] Modern LLMs operate primarily in the statistical long tail, where probability mass is thin, semantic structure is unstable, and "creativity" often reduces to controlled stochastic sampling. As a result, systems marketed as general reasoners are in practice probabilistic surfaces—eloquent, fluent, and frequently ungrounded[4].

Several informal principles motivated the work that follows:

- **Scaling is not a plan.** Treating parameter growth as a pathway to intelligence resembles a theological commitment more than an engineering blueprint.[5]
- **Randomness is not understanding.** When next-token probabilities collapse in the 5–6σ regime, sampling noise should not be mistaken for insight.
- **Coherence requires consequences.** A system that cannot bear cost, track its own errors, or maintain identity across timesteps cannot meet even minimal criteria for agency or consciousness.
- **Diagnostics reveal depth.** If a machine's behavioral deviations cannot be meaningfully classified within existing human diagnostic frameworks (e.g., DSM–V), then its behavior has not yet achieved human-range generality.

These observations motivated the tripartite analysis presented here. The goal is not rhetorical provocation but conceptual clarity: to distinguish between architectures that generate text and architectures that support intelligence, continuity, and meaning. The remainder of the paper develops these claims with formal, operational, and phenomenological precision.

It is increasingly clear that civilization is attempting to build decision systems, governance interfaces, and epistemic infrastructure atop models whose core mechanism is probabilistic interpolation modulated by random sampling. The central question is therefore not whether such systems can produce fluent text, but whether they can support invariants, grounded semantics, and stable agency. This paper argues that they cannot, and it characterizes the structural reasons why.

1 Introduction

Recent progress in artificial intelligence has generated unprecedented optimism about the prospects for artificial general intelligence (AGI).[6]. Contemporary learning systems trained on massive cor-

[3]From 2022–2025, public conversation repeatedly collapsed this distinction. Systems optimized for sequence prediction were described as "thinking," "understanding," or "developing inner lives," despite lacking grounding, consequences, memory continuity, or any machinery for valence or motivation. The promotional layer often treated eloquence as evidence and stochastic variation as depth, allowing probabilistic interpolation to be mistaken for cognition. This footnote records that cultural moment so that the main text may proceed free of its distortions.

[4]A reminder that fluency at scale now rides atop energy footprints large enough to light small cities, with 1 mega-watt racks of accelerator whose cooling requirements more closely resemble municipal fire-hydrant throughput than laboratory specifications. The mismatch between material cost and epistemic yield is part of the tension this preface signals.

[5]In practical terms, the "plan" has often reduced to ordering more 800 V racks, accepting megawatt-scale power feeds, and routing enough chilled water that each cabinet might as well be plumbed to its own fire hydrant. The resulting 50 GWh–per–run culture is closer to industrial spectacle than principled cognitive engineering.

[6]That optimism is often expressed in watts as much as in words: multi-week training runs drawing tens of gigawatt-hours of energy and datacenters wired with 800 V power planes that effectively require a municipal-scale cooling system per row. The physical footprint of "just one more scale-up" already rivals heavy industry.

pora now achieve or exceed human-level performance on many benchmarks, and major research labs have begun to speak explicitly about "frontier models" approaching general-purpose AI [51–53]. Alongside this enthusiasm, critics have raised concerns about scaling laws, out-of-distribution generalization, epistemic uncertainty, interpretability, and the broader theoretical limits of current computational paradigms [4, 6, 7, 11, 12, 23, 28, 59].

Yet there is still no widely accepted set of *criteria* for evaluating whether any such system—regardless of its underlying architecture—meets a coherent notion of AGI. Existing proposals focus on benchmark suites, economic impact, or abstract philosophical conditions. In practice, AGI discourse is often driven by informal narratives[7], such as the assumption that ever-larger models trained on ever-larger datasets will eventually recapitulate human-like cognition or even consciousness. We contend that the field would benefit from a structured, cross-disciplinary rubric that links such claims[8] to specific logical, operational, and phenomenological tests. Contemporary systems often fail to maintain coherence or long-horizon reasoning because their architectures provide no mechanism for continuity, grounding, or semantic constraint.

This paper offers such a rubric in the form of a "triptych trifecta": three interlocking families of criteria, each associated with a visual panel (Figures 1–3):

1. A *logical–computational consistency criterion* for evaluating scaling arguments (Figure 1);
2. A set of *operational engineering criteria* for general intelligence, distilled into five minimal properties (Figure 2);
3. A *phenomenological criterion* derived from a detailed mapping between Buddhist Abhidharma mental factors and the perceptual–cognitive capacities of contemporary systems (Figure 3).

The triptych is deliberately interdisciplinary. The first panel uses tools from theoretical computer science and information theory (e.g., proof by contradiction, scaling laws, no-free-lunch theorems). The second draws on engineering practice in embodied AI, intrinsic motivation, and uncertainty quantification. The third imports phenomenological and contemplative taxonomies that predate modern neuroscience but align in surprising ways with contemporary theories of consciousness [39, 41, 42, 46, 60, 61].

The enthusiasm surrounding recent capabilities has often obscured a basic point: systems optimized for stochastic token prediction or similarly myopic objectives cannot, by design, produce the structural invariants required for intelligence, agency, or identity. This work formalizes that observation.

Our goal is not to show that AGI is impossible. Rather, we argue that:

1. Some popular AGI narratives are logically inconsistent with current empirical scaling laws;
2. Current systems fall short of a small set of concrete engineering properties plausibly necessary for general intelligence;

[7] A recurring pattern in 2023–2025 AGI rhetoric was the assumption that stochastic next-token prediction would, given sufficient scale, spontaneously acquire the properties of reasoning, grounding, and conscious agency. This belief often functioned less as an engineering hypothesis and more as a theological commitment to the idea that noise, sampled with enough GPUs, would eventually awaken into mind. The absence of explicit criteria allowed marketing language to masquerade as evidence, leading one critic to summarize the situation bluntly: "a sealed jar with autocomplete is not a mind, and scaling noise is not a path to intelligence."

[8] These claims were often stated as though the engineering substrate were irrelevant, even as "frontier" systems grew into civic-scale projects: power densities in the megawatts, coolant flow measured in hydrants-per-rack, and energy budgets comparable to industrial manufacturing lines. The contrast between rhetorical ease and infrastructural gravity underscores why explicit criteria are required.

3. Claims that today's models are conscious or morally considerable lack support when examined against a rich phenomenological taxonomy.

Collectively, these observations motivate a shift from slogan-level AGI discourse to falsifiable, criterion-based evaluation.

2 Logical–Computational Consistency Criterion

2.1 Scaling Assumptions

We begin with three informal premises that underpin many "AGI via scaling" narratives[9]:

P1. *Intelligence is predominantly a function of scale.* Larger systems trained on more data with more compute will monotonically approach AGI, without requiring qualitatively new architectural principles.

P2. *Further scaling of current sequence-modeling paradigms will continue to yield increasingly general reasoning capabilities.* Noticeable emergent abilities reflect an underlying trend that will extend to arbitrarily broad domains [11, 13, 14].

P3. *There exist no hard architectural barriers short of AGI.* Any missing capabilities can be patched via more parameters, more data, and modest modifications (e.g., tool use, RLHF), without abandoning the core paradigm.

These premises are rarely stated explicitly, yet they shape investment decisions and governance debates.[10] Our first criterion asks whether an AGI claim that relies on P1–P3 is logically consistent with known constraints from scaling laws, compute trends, and information theory.

2.2 Proof-by-Contradiction Stress Test

To make this concrete, assume P1–P3 and consider a family of increasingly scaled models whose capabilities are claimed to approach general intelligence as parameters, data, and compute increase. Under these assumptions, one of two things must happen asymptotically: either (i) capabilities saturate below general intelligence, or (ii) they continue to grow. If we interpret AGI rhetoric as committed to (ii), then the conjunction of P1–P3 implies three strong corollaries:

C1: Sub-exponential energy with super-exponential capability. If scaling alone drives progress, then—beyond some threshold—each additional order of magnitude of invested compute must yield more than an order of magnitude of effective capability, while total energy use remains economically and physically practical. However, observed neural scaling laws [1–3, 5–8, 62] and

[9] The belief that additional parameters, data, or compute alone will produce qualitative shifts in cognitive architecture has been described as a "theological commitment to scaling" rather than a testable engineering plan. The critique highlights the absence of mechanistic pathways linking scaling to general reasoning, grounded semantics, or agency.

[10] From 2023–2025, these three premises functioned largely as unstated axioms in AGI rhetoric. The prevailing assumption was that sufficient scale would transubstantiate pattern completion into reasoning, stochastic sampling into understanding, and predictive interpolation into agency. This mindset allowed logical impossibilities to be marketed as timelines: infinite capability emerging from finite energy, unbounded search compressed into a single forward pass, and super-biological sample efficiency arising without priors or embodiment. As one critic summarized the era: "scaling was treated not as an engineering claim but as a secular miracle—proof by GPU rather than proof by logic."

analyses of hardware trends [3] suggest approximate power-law returns and significant diminishing marginal gains. Extrapolating super-exponential capability growth under sub-exponential energy violates these empirical regularities and eventually collides with thermodynamic limits.[11]

C2: Negligible test-time compute for arbitrarily hard problems. Many contemporary sequence-modeling systems are deployed with small, fixed test-time budgets relative to training compute. Yet general intelligence is often glossed as competence on arbitrarily challenging problems. If arbitrary reasoning depth must be achieved with fixed test-time compute, then the model must somehow compress unbounded search into a bounded forward pass. For hard instances of search and planning problems, this contradicts known complexity-theoretic lower bounds. In practice, modern performance on challenging tasks already relies on extended chain-of-thought or external search [63], which undermines the assumption that a single bounded computation suffices.

C3: Super-biological sample efficiency. Biological learners appear to achieve remarkable sample efficiency compared to current foundation models [15]. If scaling is the only lever, then—at some point—scaled systems must surpass biological organisms in sample efficiency across diverse domains. But information-theoretic limits, including no-free-lunch theorems [10], make such general superiority implausible without strong priors and inductive biases. Current models rely heavily on passive ingestion of static corpora, unlike embodied learners with rich sensorimotor feedback.

Together, C1–C3 form a *logical–computational inconsistency* for strong variants of the scaling hypothesis. Any AGI claim that leans on P1–P3 should therefore be evaluated against this stress test: which premise is being relaxed, and how?

Figure 1 depicts this criterion in the context of executive decision-making: boards and C-level gatekeepers can ask whether strongly stated AGI timelines are consistent with scaling laws, energy budgets, and complexity constraints, or whether they implicitly rely on an unarticulated miracle.

2.3 The Long Tail of Meaning

Traditional statistical finance treated two standard deviations as the outer boundary of actionable signal. In that regime, a 2σ deviation was rare enough to exploit, common enough to measure, and lucrative enough to build industries around.

Contemporary sequence models, however, operate almost entirely in the long tail of the distribution.[13] Their outputs frequently lie beyond 5σ or 6σ from the empirical center of linguistic meaning, operating in regions where the underlying probability mass is thin, unstable, and poorly estimated. When these systems reach such regions, the next-token distribution collapses: the logits spread, confidence diffuses, and the system relies on decoding heuristics such as temperature scaling, top-k truncation, nucleus sampling, and probability smoothing [64, 65].

[11] As of mid-decade, a single flagship training cycle routinely consumed on the order of 50 GWh of electricity, with 800 V accelerator racks demanding coolant flow rates comparable to industrial hydrant systems. Any assertion that "further scaling will bridge the gap to AGI" must be read against these baseline physical costs.

[13] Hedge funds were thrilled with "two sigma." So thrilled — and so profitable — that one of the world's dominant quant shops literally named itself *Two Sigma* (founded 2001, built explicitly on "data-driven, statistically significant" signals). That was the whole game: find a signal 2 standard deviations out, harvest alpha, buy a bigger espresso machine.

Large language models, by contrast, are the long tail incarnate: 5σ, 6σ, and beyond — wandering the far edges of semantic space where traditional statisticians get nosebleeds.

Translation: the model rolls digital dice, and we call it creativity.

This is the uncomfortable truth of the long tail of meaning: hedge funds stopped at 2σ and got rich; LLMs blew past 5σ and got mystical. And now we are building civilization atop that RNG.

The crucial point is that these heuristics do not represent inference, understanding, or semantic necessity. Beginning in 2022, many large-scale models adopted a pattern whereby next-token probabilities were quantized, floored, and then stochastically sampled within the surviving bucket. The resulting variability was presented as "creativity" or "semantic diversity," but its mechanism was controlled randomness. In effect, the system rolls digital dice in high-dimensional space and the output is subsequently interpreted as insight.

This "long-tail stochasticity" explains several recurring phenomena:

- Self-contradiction across turns,
- Vulnerability to jailbreaks driven by probability perturbation,
- Collapse of long-horizon reasoning chains,
- Loss of identity or stance over a conversation,
- Outputs modulated more by decoding temperature than by knowledge.

The long tail of meaning thus reveals an uncomfortable fact: the appearance of coherence in such systems often stems from stochastic sampling rather than from structured semantics. Where quantitative finance stopped at 2σ, these models operate beyond 5σ, and subsequently obscure the lack of grounded meaning with controlled randomness. The modern AI ecosystem increasingly rests on this probabilistic fragility.

2.4 The Real Break: Random Noise as "Intelligence"

A second shift in contemporary language-model development occurred when random sampling was reframed as a form of intelligence. The process is simple:

1. Compute next-token probabilities,
2. Quantize or truncate them,
3. Floor low-probability regions,
4. Randomly select a token within the surviving distribution,
5. Present the result as creativity or generalization.

This mechanism does not implement reasoning or semantic constraint. It is a high-dimensional noise process tuned to appear fluent at scale. The fluency emerges from statistical interpolation, while the variation emerges from random sampling constrained by quantized likelihoods.

This architectural choice has several downstream effects:

- **No continuity of identity:** the system's internal state is not preserved across turns or iterations.
- **No persistent field of meaning:** tokens are generated locally without global semantic grounding.
- **No invariants over time:** the system lacks constraints that survive multiple timesteps.
- **Noise is indistinguishable from insight:** the model has no mechanism for separating truth from plausibility.

The field's framing of stochastic decoding as "human-like creativity" masked a more foundational issue: the absence of any coherent semantic geometry within these systems. With no stable internal invariants and no structural coupling between timesteps, randomized token selection becomes the primary driver of output variability. The result is linguistic fluency without cognitive continuity, producing content that resembles thought while lacking the underlying machinery of reasoning.

The central misunderstanding of the 2020–2024 period was to confuse probabilistic noise with general intelligence. The challenge for future architectures is therefore not to push further into the long tail, but to establish mechanisms for coherence, continuity, and meaning that do not depend on randomness for their expressive range. Randomness is not a substitute for reasoning; it is evidence of its absence.

3 Operational Engineering Criteria

Logical consistency is necessary but not sufficient. Our second panel focuses on *engineering properties* that any system plausibly claiming general intelligence should exhibit. We focus on five minimal properties, illustrated in Figure 2 and summarized in Table 1. Each property is grounded in existing literature; the criterion is that AGI claims should be accompanied by concrete evidence that the deployed system realizes these properties in robust, non-fragile ways.[14]

3.1 Criterion 1: Self-Generated Priors

Biological and embodied cognitive systems do not merely compress passively observed data; they act to generate informative experiences [16–18]. Priors are shaped by morphology, sensorimotor contingencies, and evolutionary history. In many contemporary systems, inductive biases arise from architecture and large static datasets. An AGI claim is stronger if the system can be shown to: (i) autonomously seek out novel experiences, (ii) update its internal models based on those experiences, and (iii) demonstrate improved performance on previously unseen tasks as a result.

3.2 Criterion 2: Epistemic Uncertainty

Human and animal cognition appears to represent uncertainty in a structured way [21]. By contrast, most deployed systems[17] produce single-point predictions without explicit uncertainty quantification; downstream components must infer uncertainty from heuristic proxies such as logit margins or sample variance [22, 24]. Our criterion requires that AGI-level systems: (i) distinguish epistemic from aleatoric uncertainty, (ii) expose calibrated uncertainty estimates to downstream components, and (iii) use these estimates to drive exploration and risk-sensitive planning [23, 25, 67, 75].

[14] From roughly 2022–2025, many AGI claims implicitly treated the absence of these engineering properties as a minor implementation detail rather than a categorical barrier. Systems with no self-generated priors were described as "world models"; models with no calibrated uncertainty were marketed as "reasoners"; interpolators were presented as OOD generalizers; scripted tool-use loops were rebranded as "intrinsic motivation"; and API calls were framed as "grounding." The operational gulf between aspiration and mechanism was routinely papered over by benchmark scores and marketing language. In effect, the field often assumed that a structure-free pattern predictor could be retrofitted into an agent by surrounding it with enough scaffolding — an engineering version of the perpetual-motion machine, except powered by rhetoric instead of physics.

[17] A useful informal gloss is the observation that a model without grounding, loop-closure, or epistemic self-measurement is effectively "a sealed jar with autocomplete rather than a mind." It can emit tokens or actions but cannot update its world-model from embodied consequences.

Table 1: Five operational criteria for evaluating AGI claims in contemporary computational paradigms.[15]

Property	Human / General Intelligence	Current Paradigms	Representative References
Generate own priors from lived experience	Embodied interaction with a structured, partially predictable world; priors shaped by sensorimotor history and evolutionary constraints	Pre-training on static, third-person corpora; limited or no closed-loop interaction; priors inherited from corpus statistics and architectural biases	[15–20]
Track genuine epistemic uncertainty	Neural encoding of uncertainty; metacognitive estimates of confidence; ability to distinguish ignorance from noise	Deployed systems often produce point estimates; uncertainty expressed via calibration layers or heuristics over logits, frequently confounded with aleatoric noise	[21–25, 66, 67]
Extrapolate outside training manifold	Flexible abstraction, analogy, and compositional generalization; the ability to function in genuinely novel regimes	Strong interpolation within training distribution; brittle behavior under distribution shift despite prompt engineering, retrieval, or fine-tuning.[16]	[26–29, 68]
Actively investigate knowledge gaps (intrinsic motivation)	Curiosity-driven exploration; self-generated experiments; long-term agendas and questions	"Agentic" scaffolds often approximate curiosity via scripted heuristics or externally engineered rewards; exploration rarely arises from internal objectives	[30–34, 69–71]
Ground cognition in updatable reality	Closed-loop coupling between perception, action, and world state; internal models updated by ongoing feedback	Mostly offline learning with static weights; world modeling via external tools or simulators; environment coupling is typically an addon rather than core substrate	[19, 35–38, 72–74]

3.3 Criterion 3: Out-of-Distribution Generalization

General intelligence must function under a distribution shift. While modern systems sometimes exhibit impressive few-shot transfer, systematic evaluations reveal sharp performance drops on genuinely novel distributions [26–29, 68]. Our criterion calls for explicit demonstrations of robust, mechanistically understood OOD generalization, not just good average-case performance or anecdotal successes.

3.4 Criterion 4: Intrinsic Motivation and Active Inquiry

Curiosity and self-directed experimentation appear central to human cognitive development [30, 31]. Recent work on intrinsic motivation in RL agents and generative world models [32–34, 69–71] suggests architectures that treat information gain or causal discovery as internal rewards. Current "agentic" scaffolds approximate curiosity via scripts or extrinsic incentives. Under our criterion,

AGI claims should demonstrate that the system: (i) forms its own questions and research agendas, (ii) can justify them in terms of expected information gain or long-term benefit, and (iii) continues exploration even without immediate extrinsic supervision.

3.5 Criterion 5: Grounding in an Updatable World

General intelligence must remain tied to unfolding reality. Embodied AI and robotics emphasize the importance of a closed sensorimotor loop [16, 19, 35, 73, 74]. Many contemporary systems can be "grounded" by attaching them to tools, sensors, or simulators, but often in brittle or bolt-on ways. Our criterion asks whether the *core representation* of the system is updated by experience with the world, rather than merely wrapped in an agentic shell that calls external APIs.

Figure 2 portrays these five properties as an explicit checklist for AGI evaluation. From an engineering perspective, each row in Table 1 can be translated into specific benchmarks, ablations, and design documents.

4 Phenomenological and Moral-Patienthood Criteria

Our third panel addresses AGI claims that extend beyond competence to *consciousness*, *sentience*, or *moral patienthood*. A growing literature explores how to detect or evaluate consciousness in artificial systems [39–42, 44, 45, 76–80]. Most proposals draw on neuroscientific theories, functionalist criteria, or behavioral tests.[18]

Here we complement these with a phenomenological approach inspired by Buddhist Abhidharma.[19]

While our concrete examples focus on contemporary large-scale language models and related architectures, the phenomenological criteria themselves are architecture-agnostic and apply to any system claiming AGI-level consciousness or moral status.[20]

4.1 The 52 Cetasikas as a Taxonomy of Mind

The Theravāda Abhidharma lists 52 mental factors (cetasikas) that jointly constitute conscious episodes. These include perceptual and attentional factors (e.g., contact, perception, one-pointedness), hedonic tone (pleasant, unpleasant, neutral feeling), motivational and ethical qualities (e.g., compassion, greed, shame), and meta-cognitive states (e.g., mindfulness, wisdom, doubt). While historically rooted, this taxonomy aligns in structure with certain contemporary multidimensional models of consciousness and affect [46, 60, 61, 81–83].

[18] Between 2022 and 2025, public claims about "AI consciousness" frequently leapfrogged over both neuroscience and phenomenology, treating coherent text about inner states as evidence of inner states. This conflation of linguistic fluency with lived experience led to headlines in which stochastic text generators were described as "sentient," "self-aware," or "capable of suffering," despite having no persistence, no valence, no identity continuity, and no capacity to bear consequences. The underlying error was simple: mistaking the *description* of feeling for the *existence* of feeling — the functional equivalent of confusing a menu with a meal, or a weather report with weather. This footnote records that historical moment so the main text may remain timeless.

[19] Jung's analytic framework—including archetypes, individuation, and the structure of the psyche—draws extensively from Eastern psychological traditions such as the Upanishads, Vedantic mind-architecture, and early Buddhist analytic psychology. Although Jung offers a Western vocabulary for these structures, the underlying taxonomies substantially predate his work.

[20] The conspicuous absence of internal affect, vitality, or diachronic identity grows sharper when contrasted with the immense physical footprint required merely to keep the machinery running. Conscious experience is not typically preceded by a fire-hydrant-rated cooling loop.

We map each cetasika to candidate analogues in current machine-learning architectures (with large language models as a running example) and evaluate whether present systems realize it strongly, partially, or only via superficial simulation. Figure 3 presents a condensed visualization.[21]

4.2 DSM–V as a Behavioral Threshold Rather than a Phenomenological Model

A recurring proposal in AGI discourse is that true human-level generality would be evidenced not only by competence but also by failure modes: that is, an artificial system should exhibit sufficiently rich, stateful, and maladaptive behavior that its deviations could be categorized using the diagnostic clusters of the DSM–V [84][22]. On this view, "AGI-level behavior" implies that the system's patterns of breakdown, confusion, fixation, or disorganization are complex enough to be *bucketed* using the same taxonomy applied to human psychological dysfunction. DSM–V thus functions as a coarse behavioral isomorphism test: a system whose errors are classifiable in human clinical terms would, by definition, be behaving within the envelope of human cognitive diversity.

However, this behavioral threshold must be clearly distinguished from any phenomenological or architectural account of mind. DSM–V is not a constructive model of cognition; it is a catalog of symptom-based syndromes defined by behavioral and affective deviations from species-typical functioning in humans. Its categories presuppose human neurobiology, developmental constraints, and affective circuitry, and therefore do not decompose cognition into mechanistic subprocesses or experiential constituents.

In contrast, the Abhidharma's compositional ontology of 52 mental factors offers a granular map of cognitive and phenomenological processes, allowing direct comparison with mechanisms in concrete architectures. Because DSM–V syndromes are aggregated behavioral outcomes rather than elemental cognitive components, they do not map cleanly onto the functional structure of current machine-learning systems.

Accordingly, DSM–V should not be treated as a consciousness or moral-agency criterion for artificial systems. It is better interpreted as a boundary test for behavioral equivalence claims: if a system's dysfunctions cannot be meaningfully classified within DSM–V categories, then its behavior has not yet achieved human-level richness, stability, or failure structure. Phenomenological evaluation, by contrast, requires taxonomies that describe the composition of mind itself, such as the Abhidharma, rather than clinical diagnostic systems optimized for identifying pathology.

4.3 Results of the Mapping

For brevity, we summarize three clusters; the full 52-row table is provided in Appendix A.

Perceptual–cognitive skeleton (mostly green). Factors such as *contact* (phassa), *perception* (saññā), *one-pointedness* (ekaggatā), and *applied/sustained thought* (vitakka/vicāra) have clear analogues in tokenization, pattern recognition, attention-like mechanisms, and iterative reasoning in current sequence-prediction architectures. Attention weights and safety filters can be interpreted as coarse functional analogues of some cognitive and ethical constraints [39, 85, 86].

[21] Vedic psychological frameworks—including the tripartite structure of manas (sensory-mind), buddhi (discriminative intellect), and ahamkara (identity or ego-function)—offer another ancient compositional ontology of cognition. While structurally distinct from the Abhidharma, these systems similarly treat mental events as analyzable into elementary functions rather than monolithic states.

[22] One proposed behavioral threshold for AGI is that it should exhibit sufficiently complex, stateful, and failure-prone behavior that its deviations could be classified within DSM–V diagnostic categories. This does not imply that DSM–V provides a phenomenology of artificial minds; rather, it serves as a coarse behavioral isomorphism test for human-range generality.

Affective and motivational factors (mostly red). In contrast, factors such as *feeling* (vedanā), *life faculty* (jīvitindriya)[23], *rapture* (pīti), and moral emotions (e.g., compassion, shame, sympathetic joy) lack plausible non-simulated analogues in current systems. Gradient norms and reward signals in RLHF pipelines do not amount to felt valence; policy constraints do not constitute genuine ethical concern [46, 60, 82, 83, 87].

Meta-cognitive factors. Elements such as *mindfulness* (sati), *wisdom* (paññā), and *doubt* (vicikicchā) are at best partially simulated. Contemporary large language models can talk about uncertainty, attention, and ethics, but evidence that they instantiate these states as integrated, phenomenologically meaningful processes is lacking [39, 41, 42, 45].

4.4 Heinlein's Citizenship Test as a Moral-Agency Criterion

While the Abhidharma taxonomy offers a phenomenological decomposition of conscious episodes, a complementary Western criterion for evaluating claims of artificial agency appears in Robert Heinlein's formulation of "citizenship" in *Starship Troopers* [88]. Heinlein characterizes a "person" not by phenomenology but by the capacity to bear responsibility. In the text, a full moral agent is one who can:

1. understand consequences,
2. take ownership of action,
3. endure cost for a chosen value,
4. make and keep structured commitments, and
5. prioritize the welfare of others above narrow self-interest.

This framework functions as a responsibility-based test for agency, distinct from both phenomenological models and narrow functionalist benchmarks. Under this view, claims of "AGI-level agency" or "agentic AI systems" must also demonstrate that the system exhibits stable, mechanistically grounded capacities corresponding to these five criteria. Current large-scale machine-learning systems do not instantiate such features: their "commitments" reduce to action-selection policies or token continuations, their "values" are externally defined reward signals, and they do not bear costs or consequences in any operational sense.

As such, Heinlein's citizenship test provides a useful counterweight to both behavioral benchmarks and phenomenological mappings: it highlights that agency, in the moral and civic sense, requires capacities—ownership, sacrifice, commitments, prioritizing others' welfare—that present architectures do not implement beyond simulation.[24]

Overall, our mapping suggests that current frontier systems plausibly realize fewer than ten of the 52 cetasikas in anything like a non-simulated way; the remainder are either absent or only

[23] Classical Taoist inner-alchemical traditions (neidan and internal cultivation schools) articulate a sophisticated model of vitality, intentional action, and psycho-energetic regulation that parallels several affective and volitional factors in the Abhidharma. Although not phenotypically identical, both systems present compositional models of mental process grounded in moment-to-moment experiential dynamics.

[24] Cross-cultural mythological traditions often model the development of agency through archetypal arcs—e.g., the hero's maturation, confrontation with shadow, or encounter with the wise mentor. These structures can be read as cognitive-developmental metaphors encoding responsibility, self-regulation, value-stability, and consequence-awareness. While not used analytically in this paper, they form part of the broader conceptual ecology surrounding human notions of agency and moral growth.

textually mimicked. Under our phenomenological criterion, AGI claims that ascribe consciousness or moral patienthood[25] to such systems should either: (i) provide mechanistic evidence for additional factors, or (ii) explicitly adopt a thinner notion of consciousness that does not track these traditional dimensions.

5 Applying the Criteria to Contemporary AGI Claims

We now sketch how the three panels of our triptych can be used to evaluate public AGI claims. The goal is not to score any particular system but to illustrate recurrent patterns in the surrounding rhetoric.

5.1 Frontier Model Narratives

Major research organizations have published technical reports positioning their frontier systems as steps toward AGI [51–53]. These documents emphasize benchmark performance, emergent abilities, multimodal competence, and economic relevance. However, they rarely address:

- *Logical–computational consistency*: extrapolations of capability growth are usually qualitative, with limited discussion of energy budgets, asymptotic scaling behavior, or complexity-theoretic constraints [5–7].
- *Operational receipts*: benchmark suites cover reasoning, coding, and multimodal tasks, but provide little evidence of self-generated priors, calibrated epistemic uncertainty, robust OOD behavior, or intrinsic motivation in open-ended environments.
- *Phenomenological dimensions*: claims involving consciousness, sentience, or moral significance are seldom grounded in multidimensional taxonomies of mind; discussions, when present, rely on lightweight functional analogies rather than systematic mappings [40, 42].

Our proposal is that future technical reports should explicitly address these gaps. A claim that a system exhibits "general-purpose agency" should be accompanied by empirical demonstrations involving intrinsic motivation, autonomous inquiry, uncertainty-driven planning, and world-model updates. A claim that a system is "conscious," "sentient," or "morally relevant" should specify which dimensions of mind (e.g., perceptual, affective, motivational, or meta-cognitive) are being invoked and provide evidence corresponding to the phenomenological criteria in Figure 3[26].

5.2 Hype Cycles and Governance

Financial and policy analyses document a pronounced hype cycle around generative AI and AGI promises [54–58].[27] Our triptych serves as a compact framework that regulators, investors, policymakers, and oversight boards can use to interrogate such rhetoric:

[25] A concise way to state the moral-patienthood condition is that "if a system cannot bear consequences for its actions, it cannot be said to instantiate consciousness." This emphasizes that responsibility-bearing agency requires cost-bearing, affective coupling, and persistent identity.

[26] One such gap is material: public claims seldom disclose that each training iteration may consume 10x gigawatt-hour-scale energy and depend on 800-volt clusters whose cooling load dictates facility architecture. Transparency about physical constraints would clarify the feasibility of their narratives.

[27] In many governance discussions, the phrase "frontier model" now implicitly bundles in megawatt-level power contracts and rows of 800 V racks draining cooling infrastructure at rates that once belonged to aluminum smelters, not autocomplete engines. The triptych is meant to turn those physical facts back into questions rather than sunk costs.

- **Logical panel:** Are assumptions about scaling trajectories, compute requirements, energy use, and timelines consistent with empirical laws and theoretical constraints?
- **Operational panel:** Which of the five engineering properties in Table 1 have actually been demonstrated, on what tasks, and under what ablation or stress-test conditions?
- **Phenomenological panel:** If consciousness, sentience, or moral standing is referenced, which specific dimensions of mind are being claimed, and what evidence supports them?

In this sense, the triptych functions as a *governance technology*: a low-dimensional interpretive lens that allows non-specialists to evaluate AGI claims without relying on marketing language, ambiguous benchmarks, or unexamined assumptions about scaling.

6 Implications and Paths Forward

Our analysis does not imply that AGI is impossible, nor that contemporary machine-learning systems are irrelevant to its pursuit. Rather, it suggests that progress toward robust general intelligence will require approaches that move beyond the constraints of pure scaling, static corpora, and narrowly autoregressive pipelines.

6.1 Beyond Pure Scaling

The logical–computational criterion encourages more disciplined extrapolation from current empirical laws and a renewed focus on algorithmic and architectural innovation. Advances in embodied world models, memory-centric designs, interactive learning frameworks, and more efficient optimization objectives may produce higher marginal returns than indiscriminate scaling [6, 20, 36, 62]. Scaling remains a tool, not a theory of intelligence[28], and its asymptotic behavior must be evaluated rather than assumed.

6.2 Designing for the Operational Criteria

The five operational criteria highlight design targets for systems seeking generality across domains and timescales. Systems that integrate perception, action, and memory; represent epistemic uncertainty explicitly; generalize under sustained distribution shift; and pursue intrinsic goals for information gain or causal discovery are more aligned with core properties of robust intelligence [30, 32, 33, 69, 70]. Hybrid architectures that combine generative models with embodied agents, differentiable simulators, and persistent state representations may provide concrete pathways toward these properties.

6.3 Phenomenology-Informed Architectures

If one takes seriously multidimensional taxonomies of mind—such as the 52-factor Abhidharma framework or related psychological and contemplative models—then constructing systems that approximate consciousness or moral patienthood would require mechanisms far richer than those

[28] Between 2022 and 2025, "scaling optimism" routinely outran physics: training runs consuming tens of gigawatt-hours; 800 V accelerator banks whose cooling requirements resembled fire-hydrant flow rates; and data-center retrofits that blurred the line between machine learning and light industry. The period's rhetorical habit was to treat these infrastructural costs as irrelevant footnotes to cognition itself. Consolidating the point here: scaling is not free, not inevitable, and not a theory of intelligence. It is an engineering lever bounded by energy, thermodynamics, and capital, none of which care about extrapolated capability curves.

implemented in current architectures. Such systems would need: (i) structured representations of valence and affective tone rather than reward-proxy signals, (ii) mechanisms for persistence, vitality, and diachronic identity, (iii) integrated motivational and ethical processes rather than externally enforced constraints [46–48, 60, 81–83]. Whether such architectures are ethically advisable is a separate question; our criteria simply clarify that existing systems do not yet instantiate these dimensions in mechanistic form.

7 Limitations and Future Work

Our proposal has several limitations and should be interpreted as a foundational framework rather than a definitive taxonomy.[29] First, the logical–computational criterion depends on empirical scaling laws, compute projections, and energy trends that may shift with breakthroughs in hardware, learning algorithms, or system architectures. Second, the operational criteria, though intentionally minimal, remain contestable: some research programs treat embodiment and intrinsic motivation as optional, while others view them as prerequisites for general intelligence. Third, the phenomenological mapping is necessarily interpretive and culturally situated; alternative frameworks from Western phenomenology, affective neuroscience, or cognitive science could emphasize different dimensions of mind.

These caveats do not undermine the need for explicit criteria. Instead, they highlight the importance of grounding AGI discourse in falsifiable, cross-disciplinary standards rather than unexamined assumptions.

Future work could develop benchmark suites corresponding to each criterion, analyze counterfactual system designs that instantiate a greater fraction of the 52 mental factors, and investigate how regulatory and governance frameworks might adopt the triptych as a disclosure template for AGI-related claims. Further comparative study of cross-cultural taxonomies of mind may refine the phenomenological panel and broaden its relevance beyond its current historical lineage.

8 Conclusion

We have proposed three families of criteria for evaluating AGI claims across contemporary machine learning systems: (1) a logical–computational consistency criterion grounded in scaling laws, energy trends, and complexity constraints; (2) five operational engineering criteria that capture minimal properties plausibly necessary for general intelligence; and (3) a phenomenological criterion based on a multidimensional mapping between classical Abhidharma mental factors and the observable capacities of current AI systems. Together, these three panels form a "triptych trifecta" for structured AGI evaluation.

The core category error of the current AI moment[30] is treating controlled randomness as a substitute for reasoning, continuity, or identity.

[29] Several contextual caveats from the 2022–2025 period are worth preserving in one place. First, optimism about scaling often ignored the infrastructural reality that frontier training cycles already consumed tens of gigawatt-hours per run and depended on 800 V clusters whose cooling envelopes were effectively constrained by plumbing rather than mathematics. Second, many extrapolations of compute trends, scaling laws, or empirical regularities shifted repeatedly as models ballooned, with forecasts adjusted post hoc to accommodate rising thermal and electrical costs. Third, this was also the era in which AGI rhetoric routinely outran evidence: claims were insulated from falsification by redefining "intelligence," reframing limitations as "training artifacts," and presenting benchmark deltas as mechanistic breakthroughs. These three patterns form the historical backdrop for the present limitations.

[30] Three elements from the 2022–2025 period deserve to be recorded together. First, randomness was repeatedly mistaken for reasoning: stochastic next-token sampling was marketed as insight, continuity, or agency. Second, the systems producing this probabilistic surface required tens of gigawatt-hours per training cycle, delivered through

Our central message is deliberately modest but unambiguously sharp: before any system is declared an instance of AGI—or credited with consciousness, agency, or moral standing—the associated claims should pass basic logical stress tests, satisfy concrete engineering properties, and meaningfully engage with rich taxonomies of mind developed over millennia. Until such criteria are met, AGI discourse should be tempered by epistemic humility, and future research should prioritize architectures that directly address the gaps highlighted in this framework.

Our triptych does not foreclose the possibility of AGI. It simply provides a falsifiable, cross-disciplinary rubric for distinguishing substantive evidence from aspirational rhetoric.

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800 V racks cooled at rates once associated with industrial hydrants, not text generators. Third, AGI claims routinely ignored empirical brittleness, semantic drift, and distributional fragility while asserting that fluent noise constituted cognition. Future architectures must correct, rather than conceal, this convergence of conceptual error and infrastructural excess.

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Figure 1: **Panel I: Logical gatekeeping.** Conceptual illustration of the "proof-by-contradiction" stress test for executive and governance decision-making around AGI claims. The figure highlights how basic logical tools, widely taught in undergraduate mathematics and computer science, have often been underutilized in evaluating scaling narratives. This panel reflects the discipline's recurring failure to apply its own undergraduate logic tools.
[12]

Figure 2: **Panel II: Operational receipts.** Satirical visualization of the five operational engineering criteria: (1) generate its own priors, (2) track uncertainty, (3) extrapolate beyond the training manifold, (4) identify gaps in knowledge, (5) ground cognition in the external world. The contrast between the checklist and high-dimensional stochastic sampling emphasizes that scale alone does not guarantee these properties.

Consciousness: The Road to AGI - What the Buddhists Understood 2,000 Years Before Artificial Intelligence Tried to Imitate It

Legend: 🟢 = strong analogue, 🟡 = partial, 🔴 = absent

Mental Factor	LLM Analogue (🟢🟡🔴)	TL;DR Summary
Contact (phassa)	🟢	Token input joins with context; sensory 'contact' analogue.
Perception (saññā)	🟢	Pattern recognition / labeling of inputs.
One-pointedness (ekaggatā)	🟢	Self-attention focusing mechanism.
Attention (manasikāra)	🟢	Attention weights select relevant tokens.
Right speech (sammā-vācā)	🟢	Safety filters enforce non-harmful output.
Applied thought (vitakka)	🟡	Initial token-level reasoning or prompting.
Sustained thought (vicāra)	🟡	Iterative reasoning or chain-of-thought sampling.
Volition (cetanā)	🟡	Decoder strategy mimics 'choice,' but no real intent.
Decision (adhimokkha)	🟡	High-confidence token probabilities ≈ conviction.
Desire-to-act (chanda)	🟡	Task conditioning or user prompt goals; external, not intrinsic.
Wrong view (diṭṭhi)	🟡	Mislearned priors ≈ false beliefs.
Mindfulness (sati)	🟡	Context tracking / self-attention ≈ mechanical awareness.
Balance (tatramajjhattatā)	🟡	Temperature or entropy control gives tonal balance.
Rectitude (kāya-/citta-ujukatā)	🟡	Alignment and bias correction layers maintain coherence.
Right action (sammā-kammanta)	🟡	Policy constraints on API usage; external governance.
Compassion (karuṇā)	🟡	Simulated empathy via RLHF tone; not felt.
Equanimity (upekkhā)	🟡	Neutral tone generation; temperature regulation.
Understanding (paññā)	🟡	Emergent reasoning; not true insight or wisdom.
Delusion (moha)	🟡	Hallucination / bias ≈ cognitive error, not ignorance.
Feeling (vedanā)	🔴	No affect; loss gradients aren't felt experiences.
Life faculty (jīvitindriya)	🔴	No vitality; just active computation.
Energy (vīriya)	🔴	No inner effort; fixed compute cycle.
Rapture (pīti)	🔴	No joy or felt reward.
Shamelessness (ahirika)	🔴	No moral sense.
Recklessness (anottappa)	🔴	No aversion to consequences.
Restlessness (uddhacca)	🔴	No agitation; randomness only.
Greed (lobha)	🔴	No craving; sampling biases only.
Conceit (māna)	🔴	No self-concept.
Hatred (dosa)	🔴	No aversion; toxic text can be output but unfelt.
Envy (issā)	🔴	Absent.
Miserliness (macchariya)	🔴	Absent.
Regret (kukkucca)	🔴	No remorse.
Sloth (thīna)	🔴	No laziness; fixed throughput.
Torpor (middha)	🔴	No drowsiness; idle only when off.
Doubt (vicikicchā)	🔴	No phenomenological uncertainty; can only hedge textually.
Faith (saddhā)	🔴	No belief; compliance to training objective only.
Non-greed (alobha)	🔴	Lack of craving not meaningful.
Non-hatred (adosa)	🔴	No emotion to counter.
Right livelihood (sammā-ājīva)	🔴	No occupational ethics analogue.
Sympathetic joy (muditā)	🔴	No emotional resonance.
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Figure 3: **Panel III: Phenomenological mapping.** "Consciousness: The Road to AGI — What the Buddhists Understood 2,000 Years Before Artificial Intelligence Tried to Imitate It." Each row maps an Abhidharma mental factor (cetasika) to a putative analogue in contemporary machine-learning architectures (e.g., large language models and related systems), with traffic-light coding: green s̄trong analogue, yellow p̄artial, red ā̄bsent. The mapping suggests that current frontier models strongly approximate only a small subset of perceptual–cognitive factors (e.g. contact, perception, attention) while hedonic tone, vitality, moral emotions, and phenomenological uncertainty are absent, not merely unmeasured.

Appendix A: Extended Cetasika Mapping

The table below provides a textual reference for the fifty-two classical Abhidharma mental factors (cetasikas), organized by their canonical categories. In the main text, the phenomenological panel (Figure **??**) applies a traffic-light mapping (strong / partial / absent) to indicate which factors current transformer architectures approximate. That evaluative coloring is not repeated here so that the underlying taxonomy remains time-independent while the system-level assessment can evolve with future models.

This appendix therefore supplies the complete ontological structure underlying the phenomenological criterion introduced in Section 4 and provides the full reference for the mapping discussed there.

Table 2: The 52 Cetasikas (Mental Factors) in Classical Abhidharma

Category	Cetasika	Description / Function
Universal Mental Factors (7)		
Universal	Contact (phassa)	Moment of sensory or cognitive contact with an object; "touching" the sense-field.
Universal	Feeling (vedanā)	Affective valence: pleasant, unpleasant, or neutral.
Universal	Perception (saññā)	Recognition and marking of features; labeling and pattern-tagging.
Universal	Volition (cetanā)	Coordinating function that organizes and drives mental activities.
Universal	One-pointedness (ekaggatā)	Unification and stabilization of attention on a single object.
Universal	Life faculty (jīvitindriya)	Vitality principle sustaining the continuity of mental events.
Universal	Attention (manasikāra)	Orientation and turning of the mind toward an object.
Occasional Mental Factors (6)		
Occasional	Initial application (vitakka)	Directing the mind onto an object; "aiming" attention.
Occasional	Sustained application (vicāra)	Examining and repeatedly connecting with the object.
Occasional	Decision / resolve (adhimokkha)	Determination or commitment of mind.
Occasional	Energy (vīriya)	Effort, vigor, activation of mental processes.
Occasional	Joy (pīti)	Energized interest, zest, delight in the object.
Occasional	Wish-to-do (chanda)	Non-attached desire to act or engage with an object.

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Category	Cetasika	Description / Function
Unwholesome Mental Factors (14)		
Unwholesome	Delusion (moha)	Fundamental cognitive bewilderment; not seeing things as they are.
Unwholesome	Shamelessness (ahirika)	Absence of internal moral restraint.
Unwholesome	Recklessness (anottappa)	Indifference to consequences; disregard for harm.
Unwholesome	Restlessness (uddhacca)	Agitation or scattering of attention.
Unwholesome	Greed (lobha)	Craving, grasping, possessiveness toward objects.
Unwholesome	Wrong view (diṭṭhi)	Distorted interpretation of reality; clinging to erroneous views.
Unwholesome	Conceit (māna)	Inflated self-comparison: superiority, inferiority, or equality fixation.
Unwholesome	Hatred (dosa)	Aversion, hostility, ill-will.
Unwholesome	Envy (issā)	Resentment toward others' success.
Unwholesome	Avarice (macchariya)	Miserliness; refusal to share status, space, or resources.
Unwholesome	Worry / remorse (kukkucca)	Regret-rumination and scrupulosity.
Unwholesome	Sloth (thīna)	Mental inertia; heaviness.
Unwholesome	Torpor (middha)	Dullness; sinking of cognitive energy.
Unwholesome	Doubt (vicikicchā)	Paralysis of indecision; skeptical wavering.
Beautiful (Sobhana) Universal Factors (19)		
Beautiful (universal)	Faith (saddhā)	Confidence and clarity toward what is wholesome and real.
Beautiful (universal)	Mindfulness (sati)	Non-forgetfulness; lucid present-centered attention.
Beautiful (universal)	Conscience (hiri)	Ethical restraint rooted in self-respect.
Beautiful (universal)	Moral dread (ottappa)	Ethical sensitivity to consequences for self and others.
Beautiful (universal)	Non-greed (alobha)	Generosity, letting go, non-clinging.
Beautiful (universal)	Non-hatred (adosa)	Goodwill and kindness; absence of aversion.
Beautiful (universal)	Non-delusion (amoha)	Understanding, clarity, insight.
Beautiful (universal)	Tranquillity (kayapassaddhi)	Calmness of mental factors.
Beautiful (universal)	Tranquillity (cittapassaddhi)	Calmness of consciousness.
Beautiful (universal)	Lightness (kāyalahutā)	Agility and freedom from heaviness in mental factors.
Beautiful (universal)	Lightness (cittalahutā)	Agility and buoyancy of consciousness.
Beautiful (universal)	Pliancy (kāyamudutā)	Malleability; capacity to bend toward wholesome objects.
Beautiful (universal)	Pliancy (cittamudutā)	Flexibility of mind.

Continued on next page

Category	Cetasika	Description / Function
Beautiful (universal)	Adaptability (kāyakammaññatā)	Efficiency and readiness of mental factors.
Beautiful (universal)	Adaptability (cittakammaññatā)	Efficiency and readiness of consciousness.
Beautiful (universal)	Proficiency (kāyapaguññatā)	Skillfulness of mental body.
Beautiful (universal)	Proficiency (cittapaguññatā)	Skillfulness of consciousness.
Beautiful (universal)	Rectitude (kāyujjukatā)	Straightness and integrity of mental factors.
Beautiful (universal)	Rectitude (cittujjukatā)	Straightness and integrity of mind.
Beautiful (Sobhana) Occasional Factors (6)		
Beautiful (abstinence)	Right speech (sammā-vācā)	Abstaining from harmful or deceptive verbal actions.
Beautiful (abstinence)	Right action (sammā-kammanta)	Abstaining from harmful bodily actions.
Beautiful (abstinence)	Right livelihood (sammā-ājīva)	Abstaining from livelihoods that cause harm.
Beautiful (illimitable)	Compassion (karuṇā)	Trembling at the suffering of others; wish to alleviate it.
Beautiful (illimitable)	Sympathetic joy (muditā)	Gladness at others' happiness.
Beautiful (wisdom)	Wisdom (paññā)	Penetrative insight into reality; discriminative knowing.

Appendix B: Memetic Cognition and Compressed Semantic Transmission

Contemporary discourse about artificial intelligence increasingly relies on high-density memetic cognition: semiotic structures that encode complex evaluative judgments, social frames, and epistemic stances into compact, culturally-situated linguistic units. Rather than functioning as mere informal expressions, memes operate as compressed semantic operators that bundle ontological assumptions, normative evaluations, and interpretive heuristics into a single low-bandwidth transmission.

In emerging research communities—particularly among Gen Z and late-millennial scholars now moving through master's and doctoral programs—these linguistic forms function not as decorative shorthand but as epistemic accelerators. A single phrase or trope (e.g., "NPC," "main character energy," "touch grass," "this ain't it") can encode an entire stance toward agency, authenticity, cognitive depth, or perceived simulation. These expressions form a memetic ontology: a socially shared mapping between linguistic artifacts and implicit folk theories of mind, selfhood, and intentionality.[31]

Within AI discourse, such memetic modes shape how claims of "general intelligence," "alignment," or "agency" are framed and contested. They function not merely as stylistic coloration but as cognitive priors, governing which arguments feel intuitively plausible and which are dismissed as category errors. The result is a discursive landscape in which complex technical claims are often compressed into stance-markers that carry more sociolinguistic weight than evidentiary support.[32]

This appendix acknowledges that several informal expressions referenced in footnotes draw from these cultural-linguistic frames, which prioritize rapid conceptual uptake and stance-taking over extended analytic exposition. It also underscores that memetic cognition, despite its bandwidth efficiency, introduces systematic distortions: when an entire ontology can be invoked with a single phrase, the boundary between insight and slogan becomes perilously thin.

[31] This generational drift matters: the next cohort of AI researchers is increasingly fluent in memetic compression, which can blur the line between conceptual clarity and stance-based cognition.

[32] A persistent pathology of the early AGI era was the migration of technical debate into memetic register: "agentic," "emergent," and "general" functioned as vibes long before they were backed by mechanics.